Helix OTA — Continuation

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1. Current state

Field	Value
HEAD	`634fcb50` (constitution adopted `fc4e9b8→e60cbde`, 59 commits: carriers append §11.4.167-186+§12.12; pre-build +propagation gates 167-186/§12.12, −semgrep gate (§11.4.166 REPEALED); §11.4.109 PreToolUse guard-hook wired; ETXTBSY mutex + weasyprint export fixes — 2 commits `bb3b4189+634fcb50` pushed 4/4 FF §11.4.113). Prior test-coverage HEAD `21338527`.
Phase	**CONSTITUTION e60cbde ADOPTED (2026-07-09)** — autonomous loop (§11.4.126). NEXT actionable (parallel §11.4.103): adopt pulled nested engines token_optimizer+session_orchestrator (§11.4.141/§11.4.106); UI/UX OpenDesign refinement (§11.4.162/§11.4.170). **OPERATOR DECISION PENDING (§11.4.66/§11.4.122):** canonical frontend — `clients/ota-manager` (React 19+shadcn+Tauri) vs `dashboard` (React 18+Playwright/axe) — before sinking OpenDesign effort. Prior: **TEST-COVERAGE PROGRAM** (mandate 2026-06-23) Phase 0 DONE + Phase 1 mostly done; plan+ledger `docs/research/MASTER_TEST_COVERAGE_PLAN_20260623.md`.

Field	Value
	Cuttlefish Tier-2 A/B remains VERIFIED (F112/F55).
Terminal goal	Fully validated Helix OTA control plane driving real Android A/B updates end-to-end (protocol round-trip → payload apply → slot switch → rollback) on emulated + physical targets — **MET for the emulated Cuttlefish A/B path (F112/F55 VERIFIED); RK3588 stays control-plane-only by operator decision (§11.4.133)**

Latest session (2026-07-09) — Constitution e60cbde adoption + UI/UX/OpenDesign audit

Operator directives this session: (1) fetch+pull latest constitution submodule + nested deps, adopt all new tech/rules ASAP; (2) continue the endless autonomous loop with 3-4 parallel subagents on real evidence, no bluff; (3) answer whether Helix OTA has OpenDesign-refined production UI on all platforms.

Landed + pushed 4/4 FF (no force §11.4.113), commits `bb3b4189` (semgrep meta-test deletion, split off by a `git add` pathspec fatal) + `634fcb50` (main batch, 29 files +3485/−110): - **Constitution** `fc4e9b8→e60cbde` (59 commits); nested engines pulled (token_optimizer @c0591b5, session_orchestrator @6961c99). Parent pointer bumped. - **Carriers** CLAUDE/AGENTS/GEMINI (.md+HTML+PDF §11.4.65): appended §11.4.167-174, 176-186, §12.12 (pure append, 0 deletions). - **§11.4.166 REPEALED reconciliation** (§11.4.120, not fake-pass): removed CM-SEMGREP-WIRED gate + CM-COVENANT-114-166 propagation + `_semgrep_scan_check` in commit_all.sh + the semgrep meta-test; **added** propagation gates 167-174/176-186/§12.12 to pre_build. - **§11.4.109 anti-forgetting** wired: `.claude/settings.json` PreToolUse guard-hook (blocks force-push/sudo/host-power/raw-emulator at the tool boundary) + `docs/AGENT_GUARDRAILS.md` (preamble+checklist) + renamed guard meta-test (26/26 bluff-proof). - **Fixes:** `coverage_extra_test.go` execStubMu mutex → ETXTBSY fork+exec fd race (golang/go#22315; 20/20 -count, 10× -race clean); `guard_benchmark_baseline.sh` GUARD-SKIP when benchstat absent (regression 9/9); `export_docs.sh` +weasyprint PDF engine (pandoc 3.10 EXIT-0, genuine PDF 1.7 — a haiku delta-reviewer's NO-GO on this was **refuted by captured same-conditions evidence** per §11.4.7, see `qa-`

`results/delta_review_20260709/refutation.md`). - **UI/UX audit answer (`docs/research/ui_ux_opendesign_audit_20260709/FINDINGS.md`):** NO — Helix OTA is NOT OpenDesign-refined. TWO React frontends exist (`clients/ota-manager` shadcn/Tauri; `dashboard` hand-rolled+Playwright/axe); OpenDesign (§11.4.162) not installed in either; host-render visual-proof (§11.4.170) unmet in both. The project CLAUDE.md "§11.4.162 latent" note is stale. - Validation: pre_build PASS, inheritance PASS, regression 9/9, meta 6/6 bluff-proof, device 10× deterministic; independent review GO.

Three parallel streams COMPLETE (§11.4.147 registry: A/B/C all complete), evidence committed: - **A — new-tech engine adoption** → `docs/research/new_tech_adoption_20260709/ADOPTION_PLAN.md`. Finding (§11.4.6/§11.4.124): server has NO LLM request path today → wiring either engine now = dead code. **token_optimizer** = adopt-as-available (first use: deterministic `log_triage` over the §11.4.128 recording corpus at first AI feature); **session_orchestrator** = needs-review (its `claim+scheduler` are the §11.4.176/§11.4.119 single-owner device-claim primitive — a prototype target for parallel device testing). Both build+test GREEN. Neither operator-blocked. §11.4.141 token-efficiency wins are separate Claude-Code config. - **B — server health** → `docs/research/server_health_20260709/REMEDIATION.md`. `go build/vet/gofmt/test` all GREEN, 15/15 pkgs (forced `-count=1` non-cached). **No fix required** (zero-finding tree; correctly changed nothing per §11.4.102). Tracked gap unchanged: `internal/api/manager-dist` has no `_test.go`. - **C — OpenDesign + §11.4.170 harness** → `docs/research/ui_ux_opendesign_audit_20260709/OPENDESIGN_PLAN.md`. Recommends **`clients/ota-manager` canonical** (already has the token/dark-mode arch OpenDesign feeds; dashboard would be a rebuild; dashboard's Playwright/axe e2e is portable). §11.4.170 harness spec: Storybook 8 + addon-themes × Playwright `toHaveScreenshot()` golden-diff + Tesseract-OCR layout oracle, self-validated golden fixtures (§11.4.107(10)).

TWO PARKED OPERATOR DECISIONS (§11.4.66/§11.4.101 — documented, NOT overnight-blocked; the loop keeps progressing non-UI work): 1. **Canonical frontend** — options: (1, recommended) `clients/ota-manager` canonical + retire `dashboard` (§11.4.122 no-silent-removal ⇒ needs explicit yes); (2) ota-manager canonical, keep dashboard; (3) dashboard canonical, rebuild+retire ota-manager; (4) keep+refine both via shared token pkg. 2. **Which "OpenDesign" (§11.4.162)** — UNCONFIRMED (§11.4.6): no URL in the mandate, no repo in `.gitmodules`/our orgs. Closest evidenced candidate: "Open

Design" (open-design.ai / `nexu-io/open-design`, CSS-token system). MUST be operator-confirmed before any install (installing a guessed package = §11.4.6 violation). This is the plan's blocking step B0.

Neither frontend was modified (§11.4.122); no OpenDesign package installed (§11.4.6). The §11.4.170 host-render harness is package-independent and could be built on the recommended frontend once decision 1 lands.

NEXT (non-UI, non-blocked): resume TEST-COVERAGE PROGRAM Phase 1 remainder — the 2026-06-23 signed-pipeline-vs-live + challenges-bank evidence is still UNTRACKED and needs RE-validation on a fresh live-system boot before commit (§11.4.108, do not commit stale evidence); optionally prototype session_orchestrator as the device-claim registry (needs-review). Known tooling follow-up: `commit_all.sh --paths` fatals when a pre-staged **deletion** path is in the list (whole `git add` aborts, stages nothing) → switch to `git add -A -- $paths` or per-path add. **OPERATOR: (a) the two decisions above; (b) rotate the posted password (§11.4.10).**

Latest session (2026-06-23) — Comprehensive test-coverage + anti-bluff program (Phase 0 DONE + Phase 1 mostly done)

Continuation of the operator mandate (genuine 100% real coverage by every test type + Challenges + HelixQA vs the real running system, anti-bluff everywhere, rock-solid physical proof, no false positives). Honest framing (§11.4.6): multi-phase program — `docs/research/MASTER_TEST_COVERAGE_PLAN_20260623.md` (Rev 2) holds the plan + the **live coverage ledger**. This round drove **Phase 1 risk-descending (§11.4.132)** deep; all items below have real captured evidence committed + 4-remote synced (HEAD `21338527`), except the two explicitly marked IN FLIGHT:

- **Integration (pgx)** — store **85.5%** / rollout **83.1%** MEASURED via real `-tags integration` podman+Postgres run (`docs/qa/20260623-postgres-integration/`).
- **F-ANTIBLUFF-LIB** — `tests/lib/anti_bluff.sh ab_pass_with_evidence` per-PASS §11.4.69 helper, mutation-proven self-test + standing guard (`docs/qa/20260623-antibluff-lib/`).
- **F-CLUSTER real-system boot** — `server/deploy/system.compose.yml` + `tests/lib/boot_real_system.sh` boot the REAL ota-server + real Postgres → `/readyz`→200 + DB-backed admin JWT login (`docs/qa/`

20260623-real-system-boot/); two §11.4.102 prod fixes (startup retry-with-backoff, harness clean-slate).

- **e2e vs REAL DB** — challenge_operational **39/39 PASS** against live Postgres (DB-proof: 14 audit_logs rows, real end-of-run delete; docs/qa/20260623-e2e-live-system/).
- **Security trust-boundary 4/4 vs live** — runtime proof the **request-supplied artifact pubkey is IGNORED** (server-config key only); throwaway test secrets redacted (§11.4.10, commits 84f32a66/1d71e182). Evidence docs/qa/20260623-trust-boundary-live/.
- **Go fuzz** — 4 targets, **10.8M execs / 0 crashers**, property-based (round-trip / antisymmetry / hex-oracle); docs/qa/20260623-fuzz/.
- **HTTP load** — p99 **14.32ms** @ 5,540 req/s, zero 5xx; docs/qa/20260623-http-load-live/.
- **Chaos (§11.4.85) 4/4 survive+recover** — postgres-kill→reconnect (validates main.go retry fix, no corruption), malformed/oversized→400, 12-race idempotent register→exactly-one-device, 400-conn churn→recovers; docs/qa/20260623-chaos-live/.
- **Benchmark** — benchstat baseline registry + negation-proven guard (7 benches; thresholds calibrated on measured variance); docs/benchmarks/ + docs/qa/20260623-benchmarks/.
- **Android JaCoCo** — wired both bricks: ota-update-engine-bridge **LINE 100%** (113/113), ota-android-agent **LINE 91.18%** (json-codec branch 55.6%→**100%/100%** via 54 new tests); docs/qa/20260623-android-jacoco/ + docs/qa/20260623-json-coverage/.
- **F-METAGATES** + **propagation-gate batch** — meta-test framework wired into pre-build; **ALL 14 propagation gates now §1.1-paired** (data-driven from pre_build source → new gates auto-covered); §11.4.166 semgrep fail-open FIXED; one real §11.4.50 flake found+fixed (10/10 det). docs/qa/20260623-metagates/ + docs/qa/20260623-propagation-metagates/.
- **Independent review (§11.4.165)** — the batch passed an independent verifier that **found** + **fixed a tautological restore-integrity bluff** in tests/meta/lib_metatest.sh.
- **IN FLIGHT (not yet committed, §11.4.6):** signed-pipeline-vs-live (tests/e2e/pipeline_signed_live.sh — a local run is **15/15 PASS**: signed upload→release→deploy→rollout→device-poll-receives-signed-update vs live system, but docs/qa/20260623-signed-pipeline-live/ is UNTRACKED) and the Challenges-bank dry-run (docs/qa/20260623-challenges-bank/, UNTRACKED). The conductor lands these.

NEXT (Phase 1 remainder + Phase 2, risk-ordered): (1) commit the in-flight signed-pipeline-vs-live + challenges-bank evidence; (2) security **saturation / DDoS** (rate-limit flood vs live); (3) **on-device agent A/B** (drive the two JVM bricks on the live Cuttlefish cvd, or RK3588 when reachable); (4) **Phase 2** — wire the Go `cmd/helixqa` orchestrator + new OTA Challenges (bad-sig-rejected, request-key-ignored, telemetry-schema, rollout-staged+halt, chaos) against the live system; (5) functional §11.4.165 media/vision gate (the §11.4.163 pipeline) beyond the now-bluff-proof grep-gate layer. Suites: regression 8/8 + meta 5/5 GREEN. cvd still live on nezha. **OPERATOR: rotate the posted password (§11.4.10).**

Prior session (2026-06-23) — Comprehensive test-coverage + anti-bluff program (Phase 0 + Phase 1 start)

Operator mandate: genuine 100% real coverage by every test type + Challenges + HelixQA vs the real production system on the configured cluster, anti-bluff everywhere, rock-solid physical proof, no false positives. Honest framing (§11.4.6): this is a multi-phase program — `docs/research/MASTER_TEST_COVERAGE_PLAN_20260623.md` holds the plan + a **live coverage ledger** (updated as each item lands). Landed this round, all real captured evidence, committed + 4-remote synced:

- **Push-all** — every owned submodule + main current on all configured mirrors (constitution ×8 etc.).
- **3 research audits + master plan** (`docs/research/`): coverage-audit (Go layer mutation-proven anti-bluff; flipping the ed25519 sig-check killed 11 tests), cluster-and-system (pgx Repository IS implemented; "cluster" aspirational for OTA = thinker), challenges-helixqa-antibluff (Challenges+HelixQA powerful but unwired vs ota-server; anti-bluff on ~2-3 of 17 gates).
- **Integration coverage CAPTURED** (`docs/qa/20260623-postgres-integration/`): real `-tags integration` pgx run on nezha — store **47.9%→85.5%**, rollout **28.2%→83.1%**.
- **F-ANTIBLUFF-LIB** (`tests/lib/anti_bluff.sh`, `docs/qa/20260623-antibluff-lib/`): the §11.4.69 per-PASS `ab_pass_with_evidence` helper + mutation-proven self-test + standing guard.
- **F-CLUSTER real-system boot PROVEN** (`docs/qa/20260623-real-system-boot/`): `server/deploy/system.compose.yml` + `tests/lib/boot_real_system.sh` boot the REAL ota-server + real Postgres on thinker → `/readyz`→200 + DB-backed admin JWT login. Two §11.4.102 production fixes: server startup retry-with-backoff (was `log.Fatalf` on first pg ping — compose/k8s crash-loop), harness clean-slate `do_down` (exit-125 recreate collision).

- **e2e vs REAL DB** (`docs/qa/20260623-e2e-live-system/`): `challenge_operational` **39/39 PASS** against live Postgres (DB-proof: 14 audit_logs rows persisted, device_groups=0 end-of-run-delete real). Finding: 4/5 e2e suites self-hosting-by-design (must own the artifact pubkey to sign) → signed-pipeline-vs-live needs a caller-pubkey F-CLUSTER mode.
- **F-METAGATES** (`tests/meta/`, `docs/qa/20260623-metagates/`): meta-test framework wired into pre-build; **4 gates now bluff-proof** via paired §1.1 mutate→FAIL→restore→PASS (coverage-minimum, semgrep-wired, 2 regression guards, evidence-lib); **§11.4.166 semgrep fail-open FIXED** (semgrep on PATH → gate passes); a real flake found+fixed (§11.4.50, 10/10 det).

NEXT (Phase 1 risk-ordered, all documented in the master plan): signed-pipeline-vs-live (caller-pubkey F-CLUSTER mode) · security trust-boundary negatives + `go test -fuzz` · HTTP load/latency p50/p95/p99 (wire `tools/loadtest`) · Android JaCoCo + on-device A/B · remaining ~14 anchor-greps → functional/paired gates · Phase 2 Challenges+HelixQA Go orchestrator vs live system. Suites: regression 8/8 + meta 4/4 GREEN. cvd still live on nezha. **OPERATOR: rotate the posted password (§11.4.10).**

Prior session (2026-06-23) — Cuttlefish Tier-2 REAL Android A/B VERIFIED on nezha (F112/F55, OTA-003 closed)

TERMINAL GOAL MET (honest §11.4.6 — real captured evidence). A real ~1 GB OTA payload was applied through `update_engine` to a live Cuttlefish cvd (build 15660610, `aosp_cf_x86_64_only_phone`, Virtual A/B + verity enforcing, 15 A/B partitions) on nezha, driven autonomously as `milosvasic` over `adb -s 127.0.0.1:6520` (no host sudo for the A/B flow): `onPayloadApplicationComplete(kSuccess)` → `UPDATE_STATUS_UPDATED_NEED_REBOOT` (115 s) → reboot slot flip `_a→_b` (VAB merge `merging→none`, `_b` marked successful) → forced-bad slot `_a` (bootctl set-slot-as-unbootable + bounded 256 KB inactive-slot boot_a write, §11.4.133) rejected → device booted known-good `_b`. The OTA payload was obtained with NO credentials (androidbuildinternal pre-signed GCS URL `storage.googleapis.com`, 1003473429 B, md5 `d90870a9a6eeece3868520d7fd3f098c` — size+md5 verified before apply).

- **Evidence:** `docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md` (+ `apply_full.log`, `slot_flip.log`, `rollback.log`, `corrupt_dd.txt`, `ab_facts.txt`, …) — read-the-screen verified per §11.4.158 (REPORT §7).

- **Status:** F112 PARTIAL→VERIFIED; F55 (Tier-2 driver) OPERATOR-BLOCKED→VERIFIED (Status.md rev 21, Status_Summary rev 13; VERIFIED 25→27, PARTIAL 4→3, OPERATOR-BLOCKED 2→1).
- **Validator:** `tests/emulator/tier2_cuttlefish_ab.sh` HONEST STATUS → VERIFIED on nezha 2026-06-23; UNCONFIRMED items resolved to FACT (bootctl/update_engine_client root-only; no-creds androidbuildinternal ota-`<BID>`.zip; Virtual A/B not legacy; corrupt = set-unbootable + bounded boot_a write); new running-container `--serial`/`HELIX_CF_SERIAL` mode (topology B) added; bash -n / sh -n clean.
- **Regression guard:** `tests/regression/guard_cuttlefish_ab_proven.sh` GREEN (§11.4.135; asserts slot_flip/rollback/kSuccess evidence + validator VERIFIED header, RED on stripped proof).
- **Full journey (provenance):** curl-download-fail → resumable wget-c recovery → 27.6 GB single-stage image → slim 1.11 GB prebuilt-deb path (containers `54aa9b2`) → operator privileged launch (`cf-launch.sh`) → cvd booted → this A/B PASS. **cvd left running on nezha.**
- **Honest boundary (§11.4.3/§11.4.112/§11.4.133):** Cuttlefish is the hardware-free A/B proxy; the RK3588 boards (F113) stay control-plane-only by operator decision (native A/B is Cuttlefish-only); `bootctl/update_engine_client` are root-only on the cvd (FACT).

Prior session (2026-06-23, earlier) — Cuttlefish slim image built + launch command VERIFIED (asset-feed + `launch_cvd` proven)

Cuttlefish moved from runbook-ready to **launch-command-VERIFIED** (honest §11.4.6 — still NOT a real-A/B PASS):

1. **Slim image built + committed** — `helix-cuttlefish:slim` built rootless on `nezha` at **1.11 GB** (vs 27.6 GB single-stage from-source) via the upstream runner-prod **prebuilt-`.deb`** path (`cuttlefish-base/cuttlefish-user` **1.54.1** from `us-apt.pkg.dev/projects/android-cuttlefish-artifacts android-cuttlefish main`, NO Bazel/cargo). `cvd version 1.54.1` executes. containers submodule `54aa9b2`; parent pointer **`659c2326`**. Saved to `/tmp/cf-slim.tar` (1.03 GiB) for rootless→rootful `load`.
2. **Assets staged + integrity-verified** — nezha `~/cf-staging/`, build **15660610** `aosp_cf_x86_64_only_phone-userdebug`: `cvd-host_package.tar.gz` (898828370 B, gzip-valid) + `img.zip` (1163637538 B, unzip-valid; original curl truncation recovered via resumable `wget -c`).

3. **Runtime model FACT (§11.4.28)** — image ships modern `cvd`; `launch_cvd` is extracted at RUNTIME by the entrypoint from the host package (`CF_HOST_PKG_URL`/`CF_IMG_URL` via `file://` over the mounted `/staging`), NOT baked.
4. **PRE-VERIFY PROOF (§4.5)** — a rootless build-matched fetch-test ran the entrypoint `file://` asset-feed end-to-end: `fetching device image` (super.img + boot/init_boot/vbmeta extracted) → `fetching host package` (`./bin/launch_cvd` present) → `launching cvd via ./bin/launch_cvd` → **launch_cvd RAN**, assembled the cvd-1 config, `Launcher Build ID: 15660610`; then EXPECTED rootless `VIRTUAL_DEVICE_BOOT_FAILED run_cvd returned 10` (no /dev/kvm/bridge). Asset-feed + `launch_cvd` discovery + config assembly **PROVEN**; only the privileged boot remains. Evidence: `docs/qa/20260623-cuttlefish-launch-verified/REPORT.md`.
5. **VERIFIED operator privileged launch** (runbook §2.3): `sudo modprobe vhost_vsock` → `sudo podman load -i /tmp/cf-slim.tar` → `sudo podman run -d --name cuttlefish --privileged --network host --device /dev/kvm …vhost-vsock …vhost-net …vsock …net/tun -v /home/milosvasic/cf-staging:/staging:ro -e CF_HOST_PKG_URL=file:///staging/cvd-host_package.tar.gz -e CF_IMG_URL=file:///staging/img.zip helix-cuttlefish:slim` → `sudo podman logs -f cuttlefish`. Rootless→rootful gap closed by the `save|load` step (§11.4.161 exception — privileged run is rootful).

HONEST BOUNDARY (superseded 2026-06-23): this prior-session boundary said F112/OTA-003 was integration-pending. That has since been DONE — the operator ran the privileged launch and the agent drove `tier2_cuttlefish_ab.sh`, capturing the real A/B apply + slot-flip + auto-rollback evidence (see the latest-session block above; `docs/qa/20260623-cuttlefish-tier2-ab/`). F112/F55 are VERIFIED.

Prior session (2026-06-22, late) — distribution mechanism, infra fixes, amber onboarded, Cuttlefish runbook-ready

LATE UPDATE (2026-06-22, distribution path now fully VERIFIED end-to-end on thinker — F114): A fully-automated, non-dry-run `HELIXTRACK_REMOTE_HOST=thinker.local bash scripts/distribute_stack.sh` is now **PROVEN GREEN**: it BUILT the helixtrack-core image on `thinker` (rootless podman-compose) from the Go 1.24 Dockerfile, brought the stack up, and a fresh container reported `podman ps`: `helixtrack-core Up (healthy)` + `helixtrack-postgres Up (healthy)`, with `curl -sf http://localhost:8080/health` → 200 `{"status":"ok"}` (FailingStreak=0). Evidence `docs/qa/20260622-222645-

`distribute-thinker-FULLY-GREEN/`. The fix chain that closed the Rev-19 blockers: distribute_stack.sh (provider-preference for `podman-compose` + nested-mkdir + build-before-up + down-before-up idempotency); `containers/compose.helixtrack.yml` `/health` healthcheck (submodule `dcef56d`); HelixTrack Core Dockerfile `golang:1.24` + restored gutted source (`3c62217`/`3483699`) + `curl` in the runtime image (`d0f4bfb`). **F114 is now VERIFIED** (Status.md rev 20, Status_Summary rev 12; VERIFIED 24→25, PARTIAL 5→4). **F115 (amber docker-fallback) and F116 (setsid persistence) stay PARTIAL** — neither real-deploy/persistence run has been executed yet.

Docs + infra consolidation (honest §11.4.6 — no new PASS claimed):

1. **Cuttlefish bring-up RUNBOOK-READY (F112 / OTA-003)** — new `docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md`: the exact operator-vs-agent step split for the real Cuttlefish A/B run on nezha (which has NO passwordless sudo). **Operator** runs the 3 privileged steps — (§2.1) verify/load `vhost_vsock`/`vhost_net` + create `/dev/vsock` if absent, (§2.2) one-time group membership, (§2.3) the `sudo podman run --privileged --network host --device /dev/kvm …vhost-vsock …vhost-net …vsock …net/tun -v ~/cf-staging:/staging cuttlefish:latest`. **Agent** drives the rest — build the image (rootless), extract the staged assets (`~/cf-staging/cvd-host_package.tar.gz` 898 MB + `img.zip` 1.16 GB), `launch_cvd --daemon`, the `tier2_cuttlefish_ab.sh` A/B-slot-flip + auto-rollback validation, evidence capture to `docs/qa/<run-id>/`. Every still-`UNCONFIRMED:` item (exact device mounts, Virtual-A/B-vs-legacy, the corrupt-slot mechanism) is verified at run time, never guessed. **HONEST BOUNDARY: this is a runbook to EXECUTE — NOT a real-A/B PASS.** F112 stays PARTIAL / integration-pending until the runbook runs with captured slot-flip + rollback evidence. Built under the §11.4.161 documented exception (`CUTTLEFISH_ROOTFUL_EXCEPTION.md`).
2. **Distribution mechanism + amber onboarded (F114/F115)** — `distribute_stack.sh` (helix layer, §11.4.28 — NOT inside the generic containers submodule) deploys the HelixTrack stack over SSH + remote rootless `podman compose`. `thinker.local` = LIVE rootless-podman target; `amber.local` onboarded 2026-06-22 (SSH key installed + docker present) with a **§11.4.161 operator-authorized docker fallback** (`HELIX_ALLOW_DOCKER_FALLBACK=1`, default-OFF rootless-or-nothing, §11.4.112 documented constraint = no rootless podman on amber yet); `nezha.local` is read/import-only. Companion doc `docs/scripts/distribute_stack.md` updated (Rev 2) with the docker-fallback section.

3. **Remote-emulator full-detachment fix (F116, §11.4.144)** — `scripts/boot_android_emulator.sh` remote launch wrapped in `setsid nohup … </dev/null >log 2>&1 &` (own session + process group) so an interrupted launching SSH session no longer kills the remote emulator (closes the known robustness gap noted in Rev 5). Companion doc `docs/scripts/boot_android_emulator.md` Rev 2. Honest boundary: on-target persistence-after-reboot verification stays operator-attended.
4. **commit_all cascade-push fix (F117)** — `commit_all.sh`/`push_all.sh` portable mkdir lock + honest exit + per-remote fetch timeout; four-upstream fan-out per §2.1/§11.4.88.

Docs: 4 new Status feature rows (F114–F117, all VERIFIED), Status.md Rev 18 + Status_Summary.md Rev 10, runbook + 2 script companion docs + their html/pdf/docx exports regenerated, docs_chain features-status synced.

Operator action items (carried + new): (1) **Cuttlefish on nezha** — run the VERIFIED privileged launch block in `docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md` §2.3 (the slim image + assets are ready; only the operator's `sudo modprobe vhost_vsock` + `sudo podman load -i /tmp/cf-slim.tar` + `sudo podman run --privileged …` remains) to unblock the real Android A/B run — the agent then drives the A/B validation; (2) **amber** — install rootless podman to retire the §11.4.161 docker-fallback exception (preferred over the fallback); (3) Cuttlefish on-target persistence verification is operator-attended; (4) physical RK3588 boards remain NON-A/B (control-plane validation only).

Prior session (2026-06-22, evening) — Cuttlefish Tier-2 container path + real RK3588 control-plane validation

Three new real capabilities landed (honest §11.4.6 — boundaries stated, not overclaimed):

1. **Cuttlefish Tier-2 containerized path (F112)** — new `pkg/cuttlefish` cvd-lifecycle wrapper in the `containers` submodule (cuttlefish.go / accel.go / cleanup.go / health.go / types.go / entrypoint.sh / Containerfile). `go test -race ./pkg/cuttlefish/` = **30 PASS
    - 1 honest topology SKIP** (no Linux+KVM on this macOS host). Rootless cannot host Cuttlefish, so a narrowest-scope **rootful-privileged documented exception** is recorded — `docs/design/CUTTLEFISH_ROOTFUL_EXCEPTION.md` (§11.4.161 documented exception via §11.4.112; image build + artifact fetch stay rootless, only `launch_cvd` is privileged). **HONEST BOUNDARY:**

the container path is BUILT + unit-tested — it is NOT yet a real-A/B PASS. The real end-to-end Android `update_engine`/AVB/dm-verity A/B run is **integration-pending** on nezha Linux+KVM (assets staging; privileged `launch_cvd` operator-gated). Native A/B fidelity is this path's job once provisioned.

2. **Real RK3588 hardware control-plane validation (F113)** — evidence `docs/qa/20260622-rk3588-controlplane/REPORT.md`. Server cross-built linux/amd64, run **rootless** on nezha (uid 1000). **Device B** (Ethernet, serial `1acdceab90248933`) **PASS**: board-originated `GET /healthz` 200, `GET /api/v1/client/update` 204 (no active deployment — correct), `POST /api/v1/client/telemetry` 202, and sink-side `GET /devices/by-hardware/1acdceab90248933 → update_state=success` (the board's own telemetry mutated server state). **Device A** (Wi-Fi, `19bbb528a1dbbc4d`) honest topology **SKIP** — VPN tun1 full-tunnel / Wi-Fi AP isolation (busybox nc to both `:18080` and `:22` time out → blocked network path, captured root cause, NOT a Helix defect). **HONEST BOUNDARY: both boards are NON-A/B** (single-slot, no `update_engine`) → this validates the control plane on real hardware, NOT native A/B apply. **ZERO device state changes** (§11.4.122/§11.4.133).
3. **Distribution repoint** — container distribution targets now `thinker.local` (live) + `amber.local` (SSH-key-pending); `nezha.local` is read/import-only.

Operator action items: (1) **amber.local** — install the SSH key (`ssh-copy-id` to amber) to bring it live as a distribution target; (2) **Cuttlefish on nezha** — the privileged `launch_cvd` needs operator sudo (+ reboot + ~30 GB `fetch_cvd`) to unblock the real Android A/B run (F112 integration-pending); (3) **assets staging in progress** for the Cuttlefish Tier-2 run; (4) the **physical RK3588 boards are NON-A/B** (single-slot) — real on-device A/B fidelity is the Cuttlefish path's job, not these boards (control-plane validation only on them).

Prior session (2026-06-22, daytime) — "do everything workable" sweep

All autonomous items GREEN with real captured evidence (no bluffs): - Test sweep GREEN (server 11/11 + 4 Go submodules 3/3, `-race` 0); **gofmt fixed** (13 files); pre-build gates PASS; docs_chain in-sync. - **Local QEMU A/B OTA 6/6 GREEN** (smoke/boot/slot-switch/rollback/OTA-apply) — evidence `docs/qa/20260622T07*`. Fixed a real §11.4.115 RED-mode polarity isolation bug in `ab_rauc_verity.sh` (RED×2+GREEN×2 deterministic). - **10 server-feature recordings regenerated**, vision-verified (203 HTTP assertions) — `docs/qa/20260622-server-recordings-`

`regen/`; Status.md §11.4.153 reconciled to durable paths. - **§11.4.166 Semgrep**: propagated §11.4.160-166 into CLAUDE/AGENTS/GEMINI; tokenless local scan wired into `commit_all.sh`; constitution pin → `09d8940` (adds `docs/semgrep/TOKEN_SETUP.md`); 7 findings triaged → **0 remaining** (TLS MinVersion fix + cited suppressions). - Submodule push parity closed (`ota-update-engine-bridge`, `ota-android-agent`); GitFlic "blocker" disproven; §11.4.30 transient `qa-results/` untracked+ignored (232 files). - nezha AVD boot proven; real Android A/B = honest **operator-attended SKIP** (Cuttlefish unprovisioned — needs sudo+reboot+fetch_cvd).

Operator action items: (1) Semgrep `SEMGREP_APP_TOKEN` — follow `constitution/docs/semgrep/TOKEN_SETUP.md` (`semgrep login`) to silence the MCP hook (optional; tokenless gate already compliant); (2) provision Cuttlefish on nezha to unblock real Android A/B; (3) RK3588 board for OTA-004/F55/F56.

Known robustness gap (tracked, NOT yet fixed — §11.4.6/§11.4.123): `scripts/boot_android_emulator.sh` — the remote qemu/AVD on nezha stays tied to the launching SSH session and exits gracefully if that session ends mid-boot-wait (the nohup does not fully detach the remote process; the final SSH-tunnel/attestation step isn't reached on interrupt). Boot itself is PROVEN (`emulator-5554`, API 36, `boot_completed=1`, evidence `qa-results/20260622T071848Z-nezha-android-ab/`). Fix candidate: wrap the remote launch in `setsid nohup … </dev/null >log 2>&1 &` (or a `systemd-run --user --scope` on nezha) for true detachment — deferred because on-target persistence verification (re-boot + interrupt test, leaves remote state) is operator-attended; a blind change to the working boot path is forbidden per §11.4.1 without rock-solid proof.

Active items

ID	Title	Status	Type
OTA-003	Emulator Tier-2 — real Android A/B (update_engine/AVB/dm-verity auto-rollback)	Completed (→ Fixed.md) — VERIFIED on nezha 2026-06-23, evidence docs/qa/20260623-cuttlefish-tier2-ab/	Task

ID	Title	Status	Type
OTA-004	Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions	Operator-blocked	Task

Recently closed items

ID	Title	Status
OTA-021	HelixTrack bidirectional sync verification	Completed
OTA-020	Database migration test	Fixed
OTA-019	Build resource stats tracker	Completed
OTA-018	ApplyPort CLI + slot manager + Ed25519 verifier	Implemented
OTA-017	U-Boot corrupt-slot auto-rollback via bootcount	Implemented
OTA-016	RAUC dd-apply to inactive slot with dm-verity	Implemented
OTA-015	A/B slot switch via U-Boot BOOT_ORDER	Implemented

Full issue details: `docs/Issues.md` · `docs/Fixed.md` · `docs/Issues_Summary.md` · `docs/Fixed_Summary.md`.

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2. Server tests — all GREEN

All 13 testable packages pass (11 internal packages + chaos + stress suites):

```
ok  internal/api
ok  internal/config
ok  internal/device
ok  internal/deviceemu
ok  internal/fabric
ok  internal/health
ok  internal/rollout
ok  internal/store
ok  internal/transport
ok  tests/chaos
ok  tests/stress
```

No-test-file packages (build-only): `cmd/applyport`, `cmd/ota-device-emu`, `cmd/ota-server`, `tools/loadtest`.

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3. Emulator status

Property	Value
Host	`nezha.local` — Linux x86_64, 62 GB RAM, KVM
Image	Android API 36 (Android 16) — `CZ_API36_Phone`
Transport	ADB over SSH tunnel (reachable at `emulator-5554` from nezha)
ADB local	No Android devices connected locally (emulator is remote via SSH tunnel)
LD_LIBRARY_PATH	`/home/milosvasic/.local/lib`
Cuttlefish Tier-2	Pending AOSP guest images

The HelixTrack API is accessible from the emulator via SSH tunnel. The CZ_API36_Phone image provides the standard Android 16 AOSP stack on an x86_64 KVM host.

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4. What’s running

- **Main stream:** Emulator Tier-2 validation (remote on nezha.local)
- **Background agents:** None currently dispatched
- **Recording directory:** `$HOME/Downloads` per §11.4.158(D) (default; no project-level override)

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5. Next actions (priority-ordered)

1. **OTA-003 — DONE (VERIFIED on nezha 2026-06-23).** Real Android A/B (`update_engine` apply → slot flip `_a→_b` → auto-rollback) proven on a live Cuttlefish cvd; evidence `docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md`; §11.4.135 guard GREEN. Migrate the Issues.md entry → Fixed.md (conductor) and confirm exports.
2. **OTA-004 — Hardware unblock.** When a physical RK3588 / Orange Pi 5 Max board becomes reachable over ADB/SSH, flash and validate Tier-3 (vendor HAL, U-Boot slot-switch, real-partition dm-verity). Boards stay control-plane-only by operator decision (§11.4.133) until then.
3. **Feature-coverage video recording.** Produce §11.4.153 mandatory per-feature real-use videos confirming every server endpoint, every emulator tier, and every submodule works — with §11.4.159 window-scoped MP4 capture + §11.4.160 vision verification.
4. **Standing regression-guard suite (§11.4.135).** Ensure every closed OTA-NNN item has its §11.4.115 polarity-switch regression test registered in the suite.

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6. Binding constraints

- **Anti-bluff (§11.4 / §107):** Every PASS carries captured physical evidence per §11.4.5 / §11.4.69 / §11.4.107. Metadata-only / config-only / absence-of-error / grep-without-runtime PASS are all forbidden.
- **No-force-push (§11.4.113):** `git push --force`, `--force-with-lease`, `+<ref>` are STRICTLY FORBIDDEN. Always merge onto latest `main` and push fast-forward-only.
- **Commit via `commit_all.sh`:** All changes use the project’s canonical commit wrapper. Direct `git add`/`git commit`/`git push` are never used.

- **Four-layer fix-verification (§11.4.108):** SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE. Runtime signature is the definition of done.
- **Independent review (§11.4.142 / §11.4.125):** Every change passes an independent code-review agent before build. Review iterates to zero-finding GO per §11.4.134.
- **Multi-upstream push (§2.1):** Every commit fans out to all four upstreams (GitHub + GitLab + GitFlic + GitVerse).
- **Rootless containers (§11.4.161):** All containerized workloads use Podman in rootless mode via `vasic-digital/containers` submodule.
- **No remote CI (§11.4.156):** All CI/CD automation is DISABLED. Enforcement is through local git hooks + `pre_build_verification.sh`.

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7. Feature tracking

Feature inventory and per-row status: `docs/features/Status.md` (Rev 18, 2026-06-22). Summary companion: `docs/features/Status_Summary.md`.

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8. Fresh session start

```
git fetch --all --prune --tags
```

Then read this file (`docs/CONTINUATION.md`) and `docs/Issues.md` for the full active-item context. **The terminal goal is MET**: Cuttlefish Tier-2 REAL Android A/B is VERIFIED on nezha 2026-06-23 (OTA-003 / F112 / F55 — evidence `docs/qa/20260623-cuttlefish-tier2-ab/`). The remaining open item is OTA-004 (Tier-3 physical RK3588), which is operator-blocked on hardware (boards stay control-plane-only by operator decision, §11.4.133). The cvd is left running on nezha.